

Rishav Bikarwar

3rd Year UG, Computer Science & Engineering
Indian Institute of Technology Kanpur

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🌐 ris27hav in rishav-bikarwar

EDUCATION

Year	Degree	Institute	CPI/%
2020 - Present	B.Tech in CSE	Indian Institute of Technology Kanpur	9.0/10
2020	Class XII (CBSE)	Aklank Day Boarding Cum Residential School, Kota	95.6%
2018	Class X (CBSE)	Kendriya Vidyalaya SECL Nowrozabad, M.P.	95.4%

SCHOLASTIC ACHIEVEMENTS

- **Academic Excellence Award**, awarded for exceptional performance in academics at IIT Kanpur Feb'23
- **AIR 1094**, Joint Entrance Examination - Advanced 2020, among 160,000 qualified candidates Oct'20
- **AIR 795**, Joint Entrance Examination - Mains 2020, among approximately 1.1 million candidates Sep'20
- **AIR 627**, Kishore Vaigyanik Protsahan Yojana (KVPY) - SX Stream, conducted by IISc Bangalore Apr'20
- **National Top 300**, Indian National Chemistry Olympiad (INChO) - appeared after clearing NSEC Mar'20

PROGRAMMING ACHIEVEMENTS

- **Silver Medal**, Inter IIT Tech Meet 11.0, IIT Kanpur - DevRev's Domain Specific QA Hackathon Feb'23
- **National Top 100**, HackOn with Amazon, 2022 - Fraud Detection Challenge (*Round 4 ongoing*) July'22
- **Silver Medal**, Inter IIT Tech Meet 10.0, IIT Kharagpur - Digital Alpha's SaaS Analyzer Challenge Apr'22
- **National Top 8**, Grand Finale, Manthan, Govt. Of India - Online Fake Content Detection Challenge Dec'21
- **National Rank 2**, P&G Technical Games 2021, Data Science Case Study Challenge (awarded PPI offer) Nov'21

WORK EXPERIENCE

- **Software Development Intern** | Prutor.AI@IITK May'22 - July'22
 - Developed a **video translation algorithm** to translate English lecture videos to local languages with proper sync
 - Developed a **dashboard** for analyzing the test performance and planning study for students as well as instructors
 - Implemented **recommendation algorithm** for questions/tests based on performance and the importance of the topic
 - Worked with **NodeJS**, **ExpressJS** in the backend, **Bootstrap**, **JQuery** in the frontend and the **MongoDB** database

KEY PROJECTS

- **Digital Alpha's SaaS Analyzer** 🌐 | *Inter IIT Tech Meet 10.0* Mar'22
 - Developed a **Window Search Algorithm**, in combination with the T5 **Question-Answering model**, to extract financial metrics from SEC filings using named entity recognition & regex pattern and achieved an average accuracy of 90%
 - Utilized **LDA algorithm** to perform topic modelling on Twitter data and **FinBERT** to perform **sentiment analysis**
 - Performed **abstractive summarization** and **question generation** using the fine-tuned Pegasus & multi-task t5 models
- **MindHive** 🌐 | *CS253: Software Development - Instructor: Prof. Indranil Saha* Jan'22 - Apr'22
 - Developed a discussion **web application** similar to StackOverflow, exclusively for the IITK campus community
 - Worked with **Bootstrap** and **JQuery** for the frontend, **Django framework** in the backend and the **SQLite** database
 - Documented the web application and performed automated unit, integration and system testing using **Selenium**
- **Wikipedia Simplifier** 🌐 | *Association of Computing Activities, IIT Kanpur* Apr'21 - Aug'21
 - Mined paraphrases by scraping text from **CCNet** and comparing the L2 distances of their sentence embeddings
 - Fine-tuned the pre-trained **BART** model by applying **simplification-specific control tokens** to the mined paraphrases
- **Model Zoo** 🌐 | *Programming Club Summer Project, IIT Kanpur* June'21 - Aug'21
 - Implemented and trained the **ResNet** model on the CIFAR dataset using **PyTorch**, achieving an accuracy of 88%
 - Comprehended deep learning approaches used in **Computer Vision** tasks such as image segmentation & transformation
 - Analyzed, implemented and trained the models from **UNet**, **UNet++** and **Perceptual Losses** research papers
- **Other Miscellaneous Course Projects** Feb'22 - Apr'22
 - **SAT Solver** 🌐: Implemented **DPLL algorithm** in **C++** and applied heuristics to further improve the performance
 - **Library Management System** 🌐: Implemented the Library Management System in **C++** using the **OOPs** concepts

TECHNICAL SKILLS

- **Programming & Scripting Languages**: C, C++, Python, JavaScript, Bash, Verilog HDL
- **Tools & Libraries**: Git/GitHub, L^AT_EX, PyTorch, Nltk, Sklearn, NumPy, Matplotlib, Pandas
- **Development**: HTML, CSS, Bootstrap, jQuery, NodeJS, ExpressJS, Django, MySQL, MongoDB

RELEVANT COURSES

Data Structures & Algorithms	Fundamentals of Computing	Discrete Mathematics for CS
Software Development	Probability Theory	Linear Algebra and ODE
Computer Organization	Propositional & First Order Logic	Real Analysis and Calculus

POSITIONS OF RESPONSIBILITY AND EXTRA-CURRICULAR

- **Academic Mentor (ESC101: Fundamentals of Computing)** | *Counselling Service, IIT Kanpur* Nov'21 - Jul'22
 - Conducted academic classes, solved doubts and provided one-to-one mentorship for students learning to code
- **Gold Medal**, Football, Udghosh'21, annual sports fest organized by IITK in which 15 colleges participated Apr'22